

Ant Colony Optimization

HEURISTICS AND METAHEURISTICS FOR OPTIMIZATION AND LEARNING

10/06/2025

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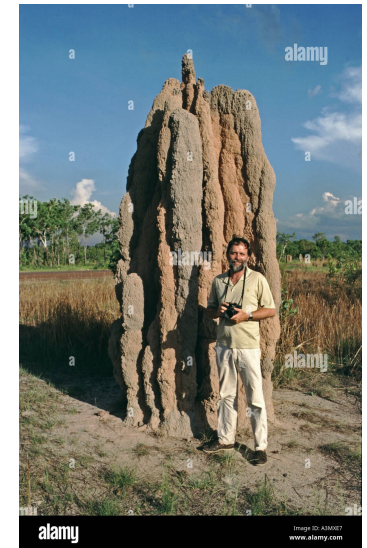
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SUMMARY

- From real ants to artificial ants
- Combinatorial optimization problems
- Swarm intelligence algorithms
- Ant colony algorithms
 - Ant system
 - Ant colony system
 - Min-max ant system
- Applications

FROM REAL ANTS TO ARTIFICIAL ANTS

- Ant colonies, and more generally social insect societies, are **distributed systems** that, in spite of the simplicity of their individuals, present a highly structured social organization.
- As a result, **they can accomplish complex tasks** that in some cases far exceed the individual capabilities of a single ant.
- The field of “**ant algorithms**” studies **models derived from the observation of real ants’ behavior** and uses these models as a **source of inspiration for the design of novel algorithms** for the solution of optimization problems.



FROM REAL ANTS TO ARTIFICIAL ANTS



- **Self-organizing principles** which allow the highly coordinated behavior of real ants can be exploited to coordinate **populations of artificial agents** that collaborate to solve computational problems.
- **Ants coordinate their activities via stigmergy**, a form of indirect communication mediated by modifications of the environment.

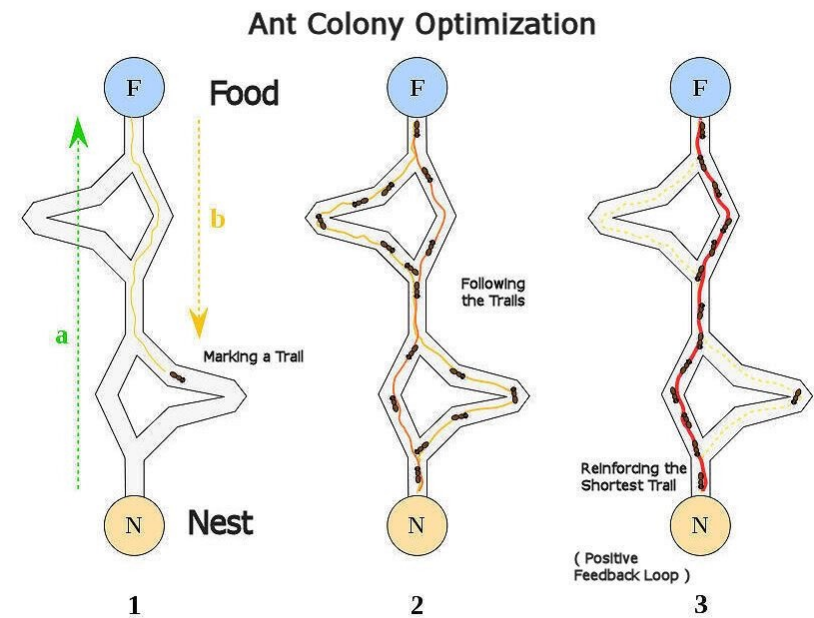
Stigmergy

“Stigmergy is a mechanism of indirect coordination between agents or actions. The principle is that the trace left in the environment by an action stimulates the performance of a next action, by the same or a different agent. In that way, subsequent actions tend to reinforce and build on each other, leading to the spontaneous emergence of coherent, apparently systematic activity.”

Wikipedia

FROM REAL ANTS TO ARTIFICIAL ANTS

- The **foraging behavior** of many ant species is **based on indirect communication** mediated by pheromones.
- **While walking** from food sources to the nest and vice versa, **ants deposit pheromones** on the ground, forming in this way a pheromone trail.
- **Ants can smell the pheromone** and they tend to **choose**, probabilistically, **paths marked by strong pheromone concentrations**.
- **Ant colony optimization,**” or ACO is **inspired by the foraging behavior of ant colonies** and targets discrete optimization problems.



http://en.wikipedia.org/wiki/Ant_colony_optimization

FROM REAL ANTS TO ARTIFICIAL ANTS

- The pheromone trail-laying and -following behavior of some ant species has been investigated in controlled **experiments** by several researchers.
- **Deneubourg and colleagues (Deneubourg, Aron, Goss, & Pasteels, 1990; Goss et al., 1989)**
 - Double bridge connecting a nest of ants and a food source.
- They ran experiments varying the ratio $r = l_l/l_s$ between the length of the two branches of the double bridge, where l_l was the length of the longer branch and l_s the length of the shorter one.

Naturwissenschaften 76, 579–581 (1989) © Springer-Verlag 1989

Self-organized Shortcuts in the Argentine Ant

S. Goss, S. Aron, J. L. Deneubourg, and J. M. Pasteels
Unit of Behavioural Ecology, C.P. 231, Université Libre de Bruxelles,
B-1050 Bruxelles

It is evident that finding the shortest route is extremely important not only for Roman road builders, thirsty rugbymen and applied mathematicians working on this very problem, but also for any animal that must move regularly between different points. How can an animal with only limited and local navigational information achieve this? Many ant and epigeic termite species illustrate this problem in the clearest possible way. The individual workers are generally less than 1 or 2 cm long, and must cover distances of 1 to 100 m between their nest and food sources. Even those that are capable, by visual or other means [1], of returning in a straight line to the nest are confronted with innumerable obstacles in their path that form a labyrinth worthy of the Minotaur. The Argentine ant *Iridomyrmex humilis* studied in this article has only a limited individual capacity for orientation [2], yet we shall see how by interacting with each other via their trail pheromone, they are capable of selecting with great reliability the shortest route between nest and food.

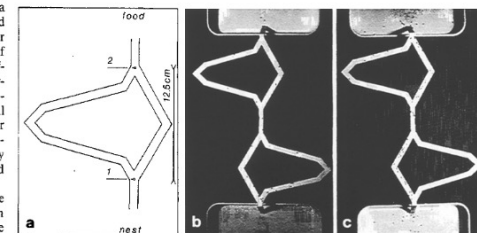
Laboratory colonies of *I. humilis* are given access to a food source in an arena linked to the nest by a bridge (Fig. 1), consisting of two identical

such that a forager going in either direction (leaving the nest or leaving the food) must choose between one or the other (at choice points 1 and 2, respectively). Each branch is at an angle of 30° to the axis of the central bridge, so that the forager has no preference for one or the other branch due to its disposition. This value of 30° has been chosen to minimize the perturbation to the forager's movement, so that on leaving one or the other branch they continue rather than double back onto

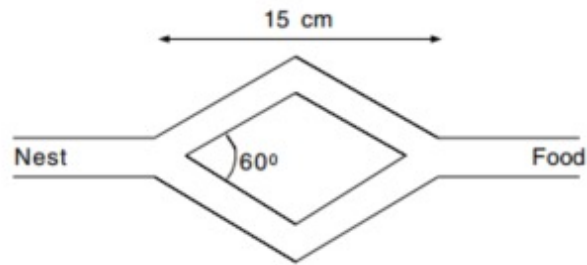
the other branch. To examine if any external bias is involved, one module's short branch is on the opposite side of the bridge as the other.

Five to 10 min after placement of the bridge, explorers have crossed it and discovered the food. Food recruitment then adds to the exploratory recruitment, and traffic on the bridge increases, the foragers at first choosing equally between the short and long branch of both modules. Abruptly, some minutes later, one branch becomes visibly preferred (Fig. 1b,c).

Knowing that *I. humilis* workers mark both leaving and returning to the nest, during food [3] and exploratory [2] recruitment, we model the behavior on the bridge as follows. ϕ ants cross the bridge in each direction per second, each laying one pheromone unit. Arriving at one of the choice points ($j = 1, 2$) of a module, each ant chooses the short or the long branch with probability p_{sj}



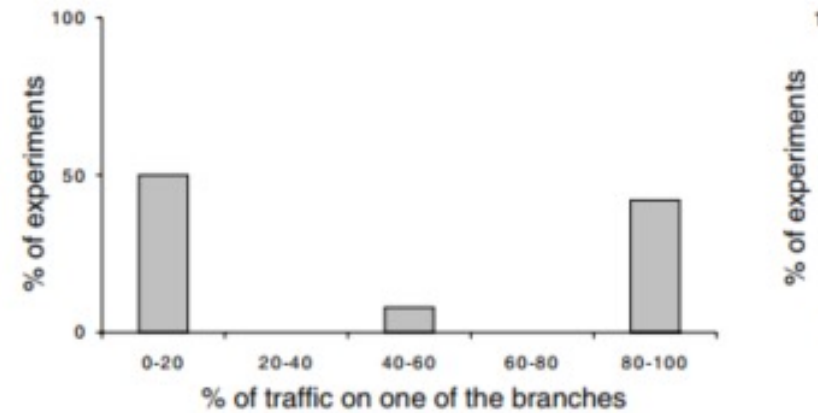
FROM REAL ANTS TO ARTIFICIAL ANTS



(a)

Figure 1.1

Experimental setup for the double bridge experiment. (a) Branches have equal length. (b) Branches have different length. Modified from Goss et al. (1989).



(a)

Figure 1.2

Results obtained with *Iridomyrmex humilis* ants in the double bridge experiment. (a) Results for the case in which the two branches have the same length ($r = 1$); in this case the ants use one branch or the other in approximately the same number of trials. (b) Results for the case in which one branch is twice as long as

FROM REAL ANTS TO ARTIFICIAL ANTS

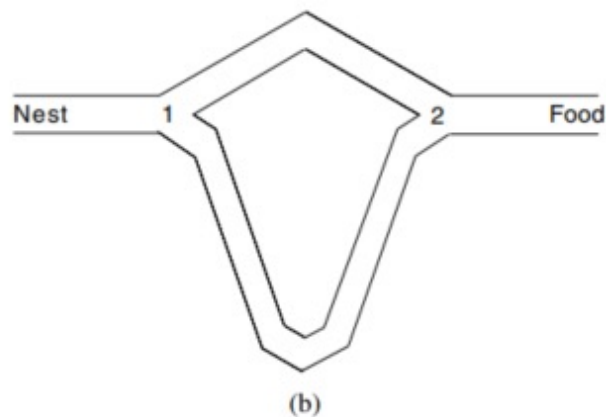


Figure 1.1

Experimental setup for the double bridge experiment. (a) Branches have equal length. (b) Branches have different length. Modified from Goss et al. (1989).

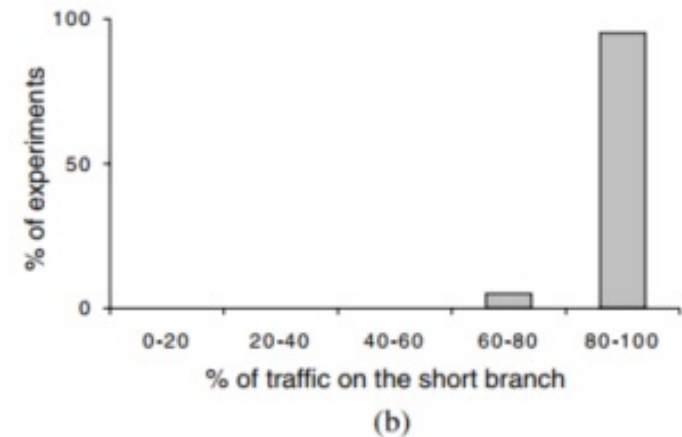
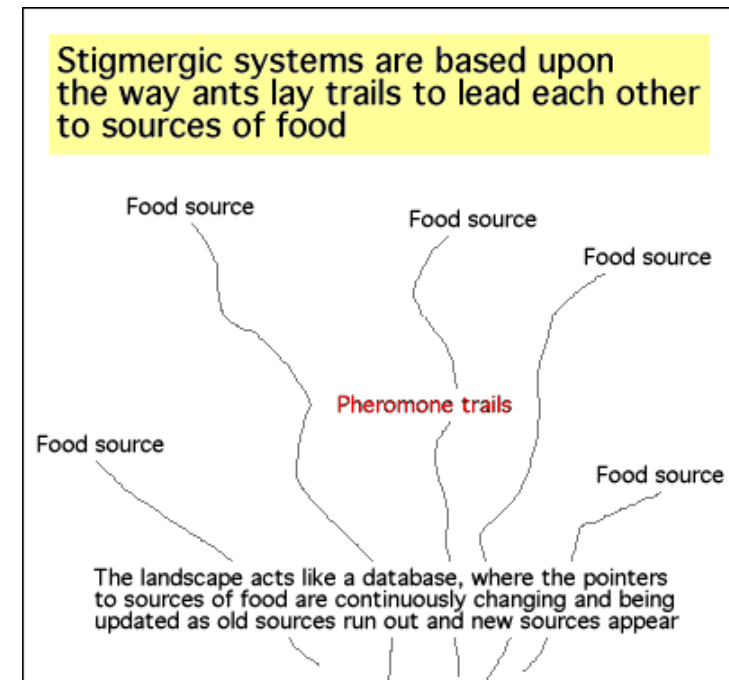


Figure 1.2

Results obtained with *Iridomyrmex humilis* ants in the double bridge experiment. (a) Results for the case in which the two branches have the same length ($r = 1$); in this case the ants use one branch or the other in approximately the same number of trials. (b) Results for the case in which one branch is twice as long as the other ($r = 2$); here in all the trials the great majority of ants chose the short branch. Modified from Goss et al. (1989).

FROM REAL ANTS TO ARTIFICIAL ANTS

- This **positive feedback** process is an **example** of:
 - **self-organizing behavior** of the ants: a **macroscopic pattern (corresponding to the convergence toward one branch)** emerges out of processes and **interactions taking place at a “microscopic” level (local interactions among the individuals of the colony)**.
 - **stigmergic communication**: ants coordinate their activities exploiting indirect communication mediated by modifications of the environment in which they move.



FROM REAL ANTS TO ARTIFICIAL ANTS

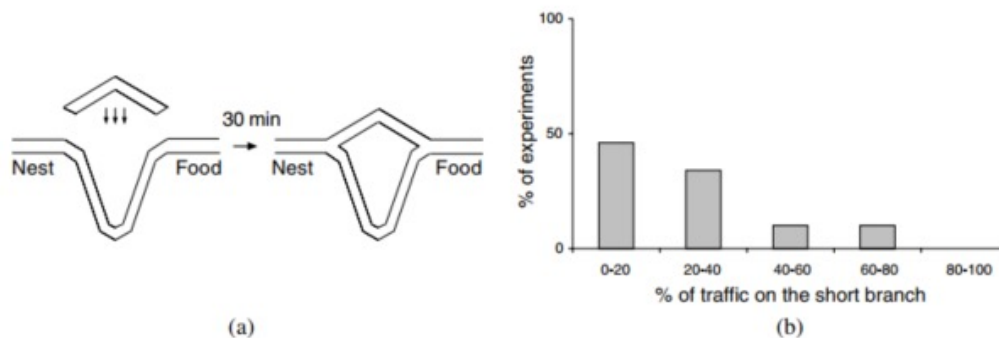


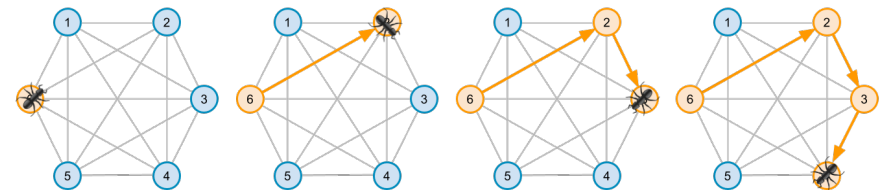
Figure 1.3

In this experiment initially only the long branch was offered to the colony. After 30 minutes, when a stable pheromone trail has formed on the only available branch, a new shorter branch is added. (a) The initial experimental setup and the new situation after 30 minutes, when the short branch was added. (b) In the great majority of the experiments, once the short branch is added the ants continue to use the long branch.

- This can be explained by **the high pheromone concentration on the long branch** and by the slow evaporation of pheromone.
- **Pheromone evaporation**, which could favor exploration of new paths, **is too slow**: the lifetime of the pheromone is comparable to the duration of a trial which means that the pheromone evaporates too slowly to allow the ant colony to “forget” the suboptimal path to which they converged so that the new and shorter one can be discovered and “learned.”

FROM REAL ANTS TO ARTIFICIAL ANTS

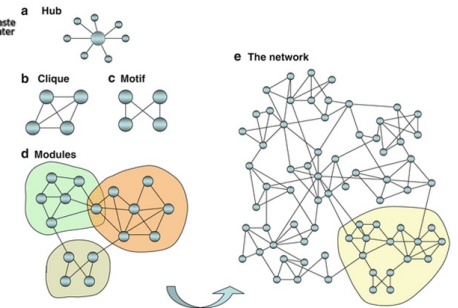
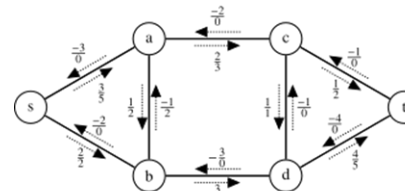
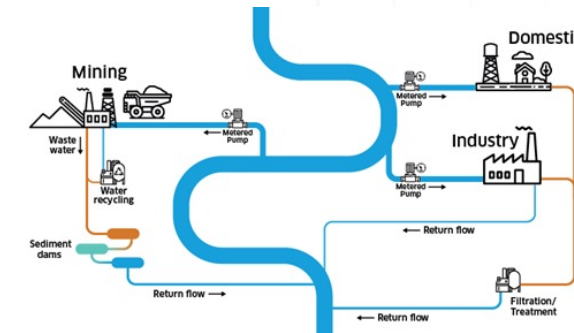
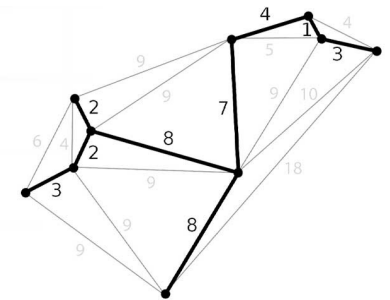
- The double bridge experiments show clearly that **ant colonies have a built-in optimization capability**: by the use of probabilistic rules based on local information they can find the shortest path between two points in their environment.
- By **taking inspiration** from the double bridge experiments, it is possible to **design artificial ants that, by moving on a graph modeling the double bridge, find the shortest path** between the two nodes corresponding to the nest and to the food source.



COMBINATORIAL OPTIMIZATION PROBLEMS

- **Combinatorial optimization problems** involve finding values for **discrete variables** such that the optimal solution with respect to a given objective function is found.
- A combinatorial optimization problem is either a **maximization** or a **minimization** problem which has associated a set of problem instances.
 - **Problem:** *general question to be answered, usually having several parameters or variables with unspecified values.*
 - **Instance:** *problem with specified values for all the parameters.*
- **Described** by a set (I, f, m, g) where:
 - I is a set of instances,
 - $f(x)$ is the set of feasible solutions for a given instance $x \in I$,
 - $m(x, y)$ is the measure of a given feasible solution y
 - g is the function to be optimized
- Find for some instance x an optimal solution for which $m(x, y) = \{g(x, y') \mid y' \in f(x)\}$

What is
Combinatorial Optimization?



(network flow, shortest path problem, transport problem, location problem, matching problem, assignment problem, routing problem)

COMBINATORIAL OPTIMIZATION PROBLEMS

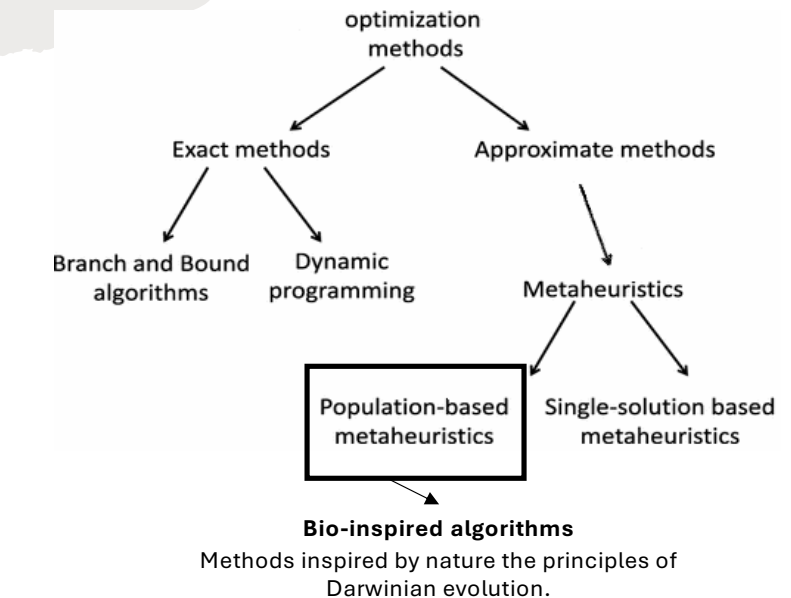
Exact methods find the optimal solution but are not always the most appropriate methods because:

Real Networks

- High dimensions.
- Non-trivial topological characteristics.
- Power-law degree distribution.
- High clustering coefficient.
- Evolve over time.

Exact methods

- High computational cost.
- Fail when applied to networks which present uncertainty, dynamism and/or incomplete information.

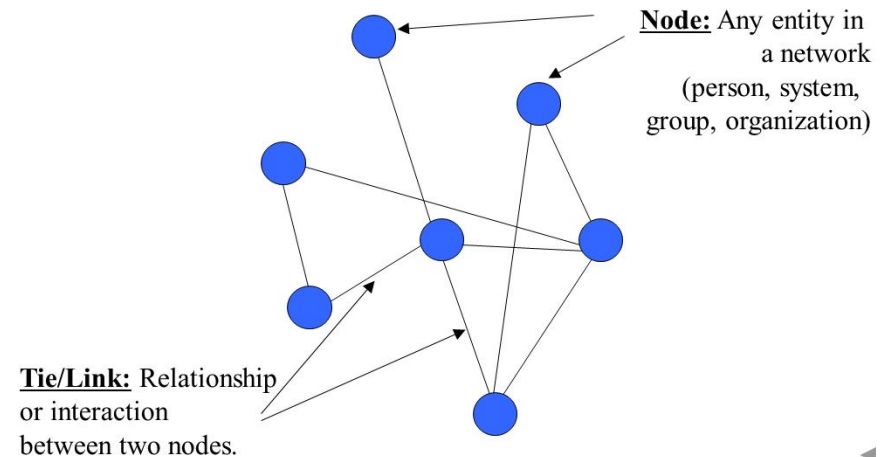


Approximate algorithms

- Don't depend on the type of problem.
- Can be used as black boxes.
- Not greedy.

COMBINATORIAL OPTIMIZATION PROBLEMS

- A **network** is defined by a pair $G(E, V)$ where $V = \{v_1, v_2, \dots, v_p\}$ is the set of **vertices** (the elements of the system) and $E = \{e_i\} \subseteq \mathbb{N} \times \mathbb{N}$ the set of **arcs** (the relations between the elements of the system), with $i = 1, 2, \dots, n$ and $e_i = \{v_l, v_m\}$ any pair of nodes.

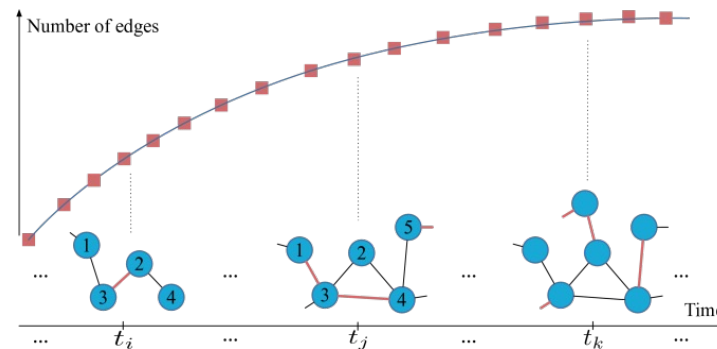
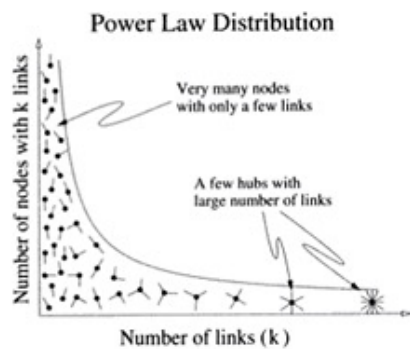


COMBINATORIAL OPTIMIZATION PROBLEMS

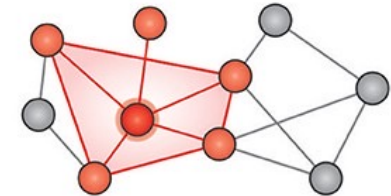
Network of real systems have **non-trivial topological characteristics**:

- Power-law degree distribution.
- High clustering coefficient.
- Evolve over time.

Dynamic Networks are defined by $G(E, V)$ where $V = \{v, t_s, t_v\}$ with v generic node, t_s and t_v the instant in which a node appears and the instant in which it disappears ($t_s \leq t_v$). $E = \{u, v, t_s, t_e\}$ is a set with $u, v \in V$ two generic nodes and t_s and t_e the instant in which a link appears between them and the instant in which the same link disappears ($t_s \leq t_e$). Nodes can appear or disappear and the links damaged or repaired. The state of the network changes over time¹.



Clustering coefficient

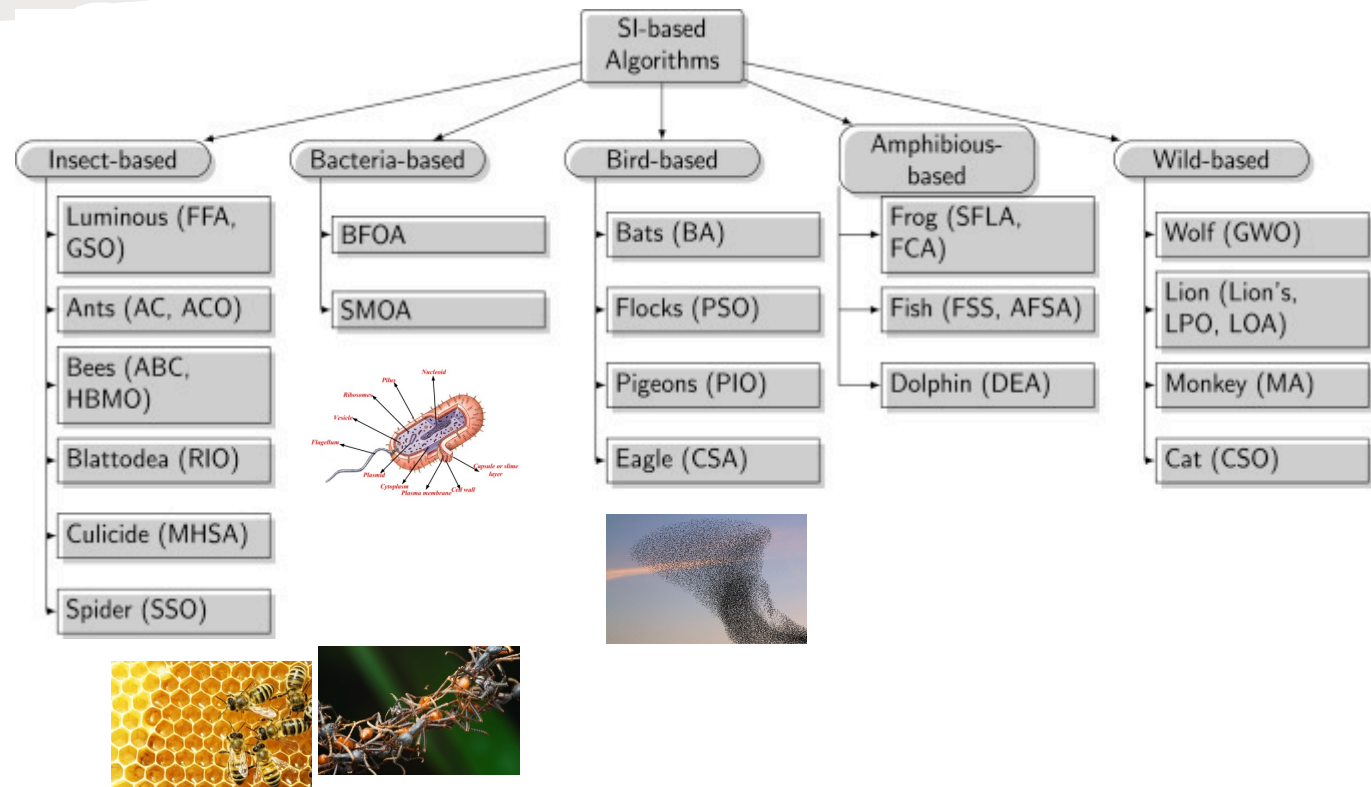


1. Joakim Skarding, Bogdan Gabrys, and Katarzyna Musial. Foundations and modelling of dynamic networks using dynamic graph neural networks: A survey. 052020.

SWARM INTELLIGENCE ALGORITHMS

The **swarm intelligence** is defined as the *collective behaviour* of decentralized and self-organized systems in which the group manifests a form of "intelligence" greater than any element of it alone.

- The **swarm can solve complex problems** that a single individual with simple abilities cannot solve.
- The swarm is **made up of several elements** that can get lost or damaged without affecting the performance of the entire group.
- **Individuals perceive information only locally**, perform simple actions, have little or no memory, and do not know what the overall state of the system is or its purpose².



ANT COLONY ALGORITHMS

Many special cases of the ACO metaheuristic have been proposed. The three most successful ones are:

- **Ant System (AS)** developed by Marco Dorigo (Italy) in his PhD thesis in 1992
- **Min-Max Ant System (MMAS):** developed by Hoos and Stützle in 1996
- **Ant Colony System (ACS):** developed by Gambardella-Dorigo in 1997

ANT SYSTEM

Proportional transition rule

$$p_{ij}^k(t) = \begin{cases} \frac{\tau_{ij}(t)^\alpha \cdot \eta_{ij}^\beta}{\sum_{l \in J_i^k} \tau_{il}(t)^\alpha \cdot \eta_{il}^\beta} & \text{if } j \in J_i^k \\ 0 & \text{if } j \notin J_i^k \end{cases}$$

- τ_{ij} pheromone intensity the path (i,j),
- α parameter that controls the importance of τ_{ij} ,
- η_{ij} visibility of the node j,
- β parameter that controls the importance η_{ij} ,
- J_i^k set of possible movements of the ant.

Reinforcement rule

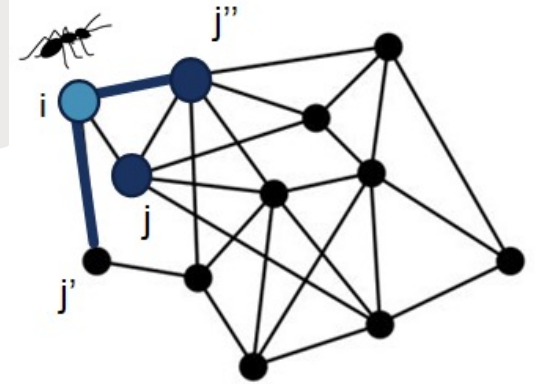
$$\Delta\tau_{ij}^k(t) = \begin{cases} \frac{Q}{L^k(t)} & \text{if } (i,j) \in T^k(t) \\ 0 & \text{if } (i,j) \notin T^k(t) \end{cases}$$

- L^k length of the path,
- T^k total path made by the ant
- Q parameter.

Global updating rule

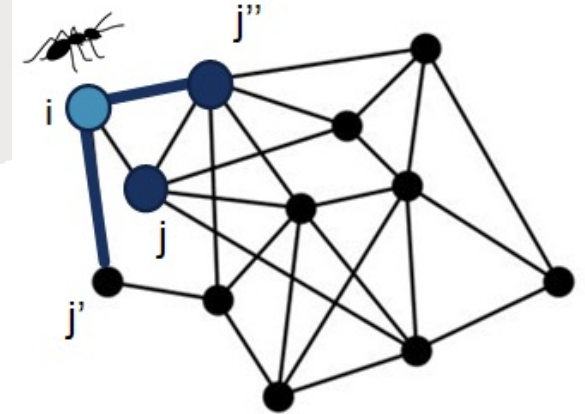
$$\tau_{ij}(t+1) = (1 - \rho) \tau_{ij}(t) + \sum_{k=1}^m \Delta\tau_{ij}^k(t)$$

- τ_{ij} pheromone intensity the path (i,j),
- $0 < \rho < 1$ pheromone evaporation rate,
- $\Delta\tau_{ij}^k$ deposited pheromone,
- m number of ants.



MIN-MAX ANT SYSTEM

- Only the global best tour or the iteration best tour are allowed to add pheromone to its trail.
- To avoid stagnation of the search algorithm, the range of possible pheromone amounts on each trail is limited to an interval $[\tau_{\max}, \tau_{\min}]$.
- All edges are initialized to $\tau_{\max}$ to force a higher exploration of solutions.
- The trails are reinitialized to $\tau_{\max}$ when nearing stagnation



Reinforcement rule

$$\Delta\tau_{ij}^{best}(t) = \begin{cases} \frac{Q}{L^{best}(t)} & \text{if } (i,j) \in T^k(t) \\ 0 & \text{if } (i,j) \notin T^k(t) \end{cases}$$

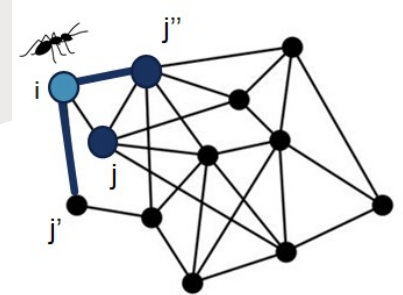
L^{best} can be set to the length of the best tour found in the current iteration or the best solution found since the start of the algorithm

Global updating rule

$$\tau_{ij}(t+1) = (1 - \rho) \tau_{ij}(t) + \rho \cdot \Delta\tau_{ij}^{best}(t)$$

$\Delta\tau_{ij}^{best}(t)$: only the best ant is allowed to update the trails

ANT COLONY SYSTEM



Pseudorandom Proportional rule

$$j = \begin{cases} \underset{AS}{\operatorname{argmax}}_{j \in J_i^k} \{ \tau_{ij} [\eta_{ij}]^\beta \} & \text{if } q \leq q_0 \\ \text{otherwise} & \end{cases}$$

Reinforcement rule

$$\Delta \tau_{ij}^{best}(t) = \begin{cases} \frac{Q}{L^{best}(t)} & \text{if } (i, j) \in T^k(t) \\ 0 & \text{if } (i, j) \notin T^k(t) \end{cases}$$

Local updating rule

$$\tau_{ij}(t+1) = (1 - \varphi) \tau_{ij}(t) + \varphi \cdot \tau_{ij}(0)$$

Global updating rule

$$\tau_{ij}(t+1) = (1 - \rho) \tau_{ij}(t) + \rho \cdot \Delta \tau_{ij}^{best}(t)$$

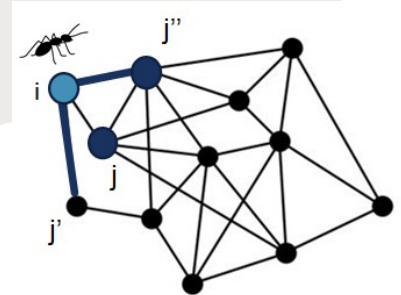
- The probability for an ant to move from a node i to a node j depends on a random variable q uniformly distributed over $[0; 1]$, and a parameter q_0 .
- If $q \leq q_0$, then, among the feasible components, the component that maximizes the product $\tau_{ij} [\eta_{ij}]^\beta$ is chosen, otherwise the same equation as in Ant System is used.
- This rule favors exploitation of pheromone information

- L^{best} can be set to the length of the best tour found in the current iteration or the best solution found since the start of the algorithm

- ants change the pheromone level of the edges they are selecting by applying a local pheromone updating rule;

- $\Delta \tau_{ij}^{best}(t)$: only the best ant is allowed to update the trails

ANT COLONY SYSTEM



Pseudorandom Proportional rule

$$j = \begin{cases} \underset{AS}{\operatorname{argmax}}_{j \in J_i^k} \{ \tau_{ij} [\eta_{ij}]^\beta \} & \text{if } q \leq q_0 \\ \text{otherwise} & \end{cases}$$

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Local updating rule

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Global updating rule

$$\tau_{ij}(t+1) = (1 - \rho) \tau_{ij}(t) + \rho \cdot \Delta \tau_{ij}^{best}(t)$$

- The probability for an ant to move from a node i to a node j depends on a random variable q uniformly distributed over $[0; 1]$, and a parameter q_0 .
- If $q \leq q_0$, then, among the feasible components, the component that maximizes the product $\tau_{ij} [\eta_{ij}]^\beta$ is chosen, otherwise the same equation as in Ant System is used.
- This rule favors exploitation of pheromone information
- L^{best} can be set to the length of the best tour found in the current iteration or the best solution found since the start of the algorithm

Aspect	Local Updating Rule	Global Updating Rule
When applied	During solution construction	At the end of each iteration
Who updates	Every ant	Only the best ant
Effect	Encourage exploration	Encourage exploitation
Main purpose	Exploration / Diversification	Exploitation / Convergence

ACO ALGORITHMS

Procedure of ACS Algorithm:

Begin

Initialize

While stopping criterion not satisfied **do**

Position each ant in a starting node

Repeat

For each ant **do**

Choose next node by applying the state transition rule

Apply step by step pheromone update

End for

Until every ant has built a solution

Update best solution

Apply offline pheromone update

End While

End

ACO ALGORITHMS

ADVANTAGES

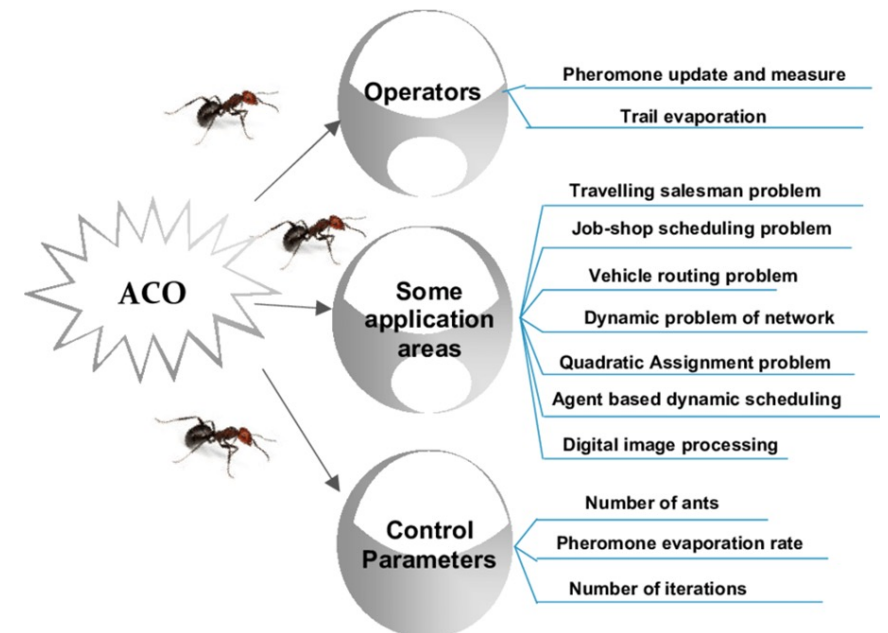
- Positive Feedback accounts for rapid discovery of good solutions
- Efficient for Traveling Salesman Problem and similar problems
- Can be used in dynamic applications (adapts to changes such as new distances, etc)

DISADVANTAGES

- Probability distribution changes by iteration
- Research is experimental rather than theoretical
- Time to convergence uncertain (but convergence is guaranteed!)

PERFORMANCES

- Algorithm found best solutions on small problems (30 city)
- On larger problems converged to good solutions but not the best
- On “static” problems like TSP hard to beat specialist algorithms



APPLICATIONS

Table 2.1
Current applications of ACO algorithms listed according to problem types and chronologically

Problem type	Problem name	Main references
Routing	Traveling salesman	Dorigo, Maniezzo, & Colomi (1991a,b, 1996) Dorigo (1992) Gambardella & Dorigo (1995) Dorigo & Gambardella (1997a,b) Stützle & Hoos (1997, 2000) Bullnheimer, Hartl, & Strauss (1999c) Cordón, de Viana, Herrera, & Morena (2000)
	Vehicle routing	Bullnheimer, Hartl, & Strauss (1999a,b) Gambardella, Taillard, & Agazzi (1999) Reimann, Stummer, & Doerner (2002)
	Sequential ordering	Gambardella & Dorigo (1997, 2000)
	Quadratic assignment	Maniezzo, Colomi, & Dorigo (1994) Stützle (1997b) Maniezzo & Colomi (1999) Maniezzo (1999) Stützle & Hoos (2000)
Assignment	Graph coloring	Costa & Hertz (1997)
	Generalized assignment	Lourenço & Serra (1998, 2002)
	Frequency assignment	Maniezzo & Carbonaro (2000)
	University course timetabling	Socha, Knowles, & Sampels (2002) Socha, Sampels, & Manfrin (2003)
Scheduling	Job shop	Colomi, Dorigo, Maniezzo, & Trubian (1994)
	Open shop	Pfahringier (1996)
	Flow shop	Stützle (1998a)
	Total tardiness	Bauer, Bullnheimer, Hartl, & Strauss (2000)
Subset	Total weighted tardiness	den Besten, Stützle, & Dorigo (2000) Merkle & Middendorf (2000, 2003a) Gagné, Price, & Gravel (2002)
	Project scheduling	Merkle, Middendorf, & Schmeck (2000a, 2002)
	Group shop	Blum (2002a, 2003a)
	Multiple knapsack	Leguizamón & Michalewicz (1999)
Subset	Max independent set	Leguizamón & Michalewicz (2000)
	Redundancy allocation	Liang & Smith (1999)
	Set covering	Leguizamón & Michalewicz (2000) Hadji, Rahoual, Talbi, & Bachelet (2000)
	Weight constrained graph tree partition	Cordone & Maffioli (2001)
Subset	Arc-weighted l -cardinality tree	Blum & Blesa (2003)
	Maximum clique	Fenet & Solnon (2003)

Table 2.1
(continued)

Problem type	Problem name	Main references
Other	Shortest common supersequence	Michel & Middendorf (1998, 1999)
	Constraint satisfaction	Solnon (2000, 2002)
	2D-HP protein folding	Shmygelska, Aguirre-Hernández, & Hoos (2002)
	Bin packing	Levine & Ducatelle (2003)
Machine learning	Classification rules	Parpinelli, Lopes, & Freitas (2002b)
	Bayesian networks	de Campos, Gámez, & Puerta (2002b)
	Fuzzy systems	Casillas, Cordón, & Herrera (2000)
Network routing	Connection-oriented network routing	Schoonderwoerd, Holland, Bruten, & Rothkrantz (1996) Schoonderwoerd, Holland, & Bruten (1997) White, Pagurek, & Oppacher (1998) Di Caro & Dorigo (1998d) Bonabeau, Henavy, Guérin, Snyers, Kuntz, & Theraulaz (1998)
	Connectionless network routing	Di Caro & Dorigo (1997, 1998c,f) Subramanian, Druschel, & Chen (1997) Heusse, Snyers, Guérin, & Kuntz (1998) van der Put (1998)
	Optical network routing	Navarro Varela, & Sinclair (1999)

- Traveling Salesman
- Assignment problems
 - Graph Coloring
- Scheduling

TSP

Starting from his hometown, a **salesman wants to find a shortest tour that takes him through a given set of customer cities and then back home, visiting each customer city exactly once**. The goal in TSP is to find a minimum length Hamiltonian circuit of the graph.

The general algorithm is based on a set of ants, each making one of the possible round-trips along the cities. At each stage, the ant chooses to move from one city to another according to some rules:

- 1) It must **visit each city exactly once**;
- 2) A **distant city** has **less chance** of being chosen (the visibility);
- 3) The **more intense the pheromone** trail laid out on an edge between two cities, the **greater the probability** that that edge will be chosen;
- 4) Having completed its journey, the ant deposits **more pheromones** on all edges it traversed, **if the journey is short**;
- 5) After each iteration, trails of **pheromones evaporate**.



ASSIGNMENTS PROBLEMS

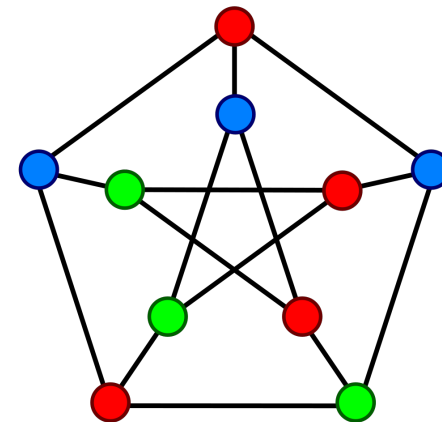
GRAPH COLORING

The task in assignment problems is to **assign a set of items** (objects, activities, etc.) **to a given number of resources** (locations, agents, etc.) **subject to some constraints**. Assignments can be represented as a mapping from a set I to a set J , and the objective function to minimize is a function of the assignments done.

Given an undirected graph $G = (N, A)$, the goal in the GCP is to **find the minimum number of colors to assign to nodes such that no pair of adjacent nodes is assigned the same color**.

In their ACO algorithm, Costa and Hertz **use pheromones to indicate the desirability of assigning the same color to two nodes**.

For ants' solution construction, they adapted the heuristics used in two constructive algorithms.



SCHEDULING PROBLEMS

Scheduling problems involve **solving for the optimal schedule** under various objectives, different machine environments and characteristics of the jobs.

In case jobs have to be processed on more than one machine, the task to be performed on a machine for completing a job is called an operation. Generally, the machine-scheduling models consider

- the processing times of all jobs and operations is fixed and known;
- the processing of jobs and operations cannot be interrupted (scheduling without preemption).

Scheduling problems play a central role in ACO research, and many different types of scheduling problems have been attacked with ACO algorithms. **The performance**, however, **varies across problems**.

